

Exercise C1

Q1. Which of the following pieces of data does NOT uniquely determine an acute-angled triangle ABC

(R being the radius of the circumcircle)?

- (A) $a, \sin A, \sin B$ (B) a, b, c (C) $a, \sin B, R$ (D) $a, \sin A, R$

Ans. D

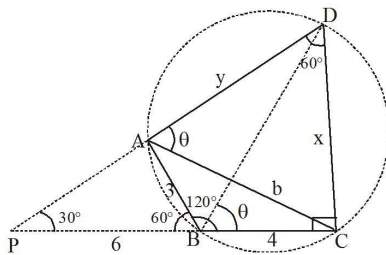
Sol. -

Q2. In a triangle ABC, $\angle ABC = 120^\circ$, $AB = 3$ and $BC = 4$. If perpendicular constructed on the side AB at A and to the side BC at C meets at D then CD is equal to

- (A) 3 (B) $\frac{8\sqrt{3}}{3}$ (C) 5 (D) $\frac{10\sqrt{3}}{3}$

Ans. D

Sol.



Extend CB and DA to meet at P.

note that $\triangle PCD$ is right angle as shown

$$\text{now } \tan 30^\circ = \frac{x}{10}$$

[PAB is $30^\circ - 60^\circ - 90^\circ$ triangle, hence $PB = (2)(3) = 6$]

$$\therefore x = \frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3}$$

Q3. If in a triangle $\sin A : \sin C = \sin(A - B) : \sin(B - C)$ then a^2, b^2, c^2

- (A) are in A.P. (B) are in G.P. (C) are in H.P. (D) none of these

Ans. A

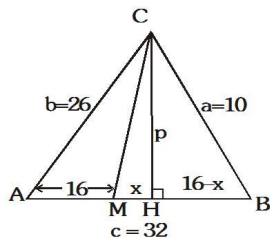
Sol. -

Q4. In a triangle ABC, CH and CM are the lengths of the altitude and median to the base AB. If $a = 10$, $b = 26$, $c = 32$ then length (HM)

- (A) 5 (B) 7 (C) 9 (D) none

Ans. C

Sol.



$$\text{In } \triangle AHC \quad p^2 + (16+x)^2 = (26)^2 \dots\dots\dots(i)$$

$$\text{In } \triangle BHC \quad p^2 + (16-x)^2 = (10)^2 \dots\dots\dots(ii)$$

$$\Rightarrow (16+x)^2 - (16-x)^2 = (26)^2 - (10)^2$$

$$\Rightarrow (32)(2x) = (36)(16)$$

$$\Rightarrow x = 9$$

Q5. In $\triangle ABC$ if $a = 8$, $b = 9$, $c = 10$, then the value of $\frac{\tan C}{\sin B}$ is

- (A) $\frac{32}{9}$ (B) $\frac{24}{7}$ (C) $\frac{21}{4}$ (D) $\frac{18}{5}$

Ans. A

$$\text{Sol. } \frac{\tan C}{\sin B} = \frac{\sin C}{\sin B} \cdot \frac{1}{\cos C}$$

$$\text{Now using sine law } \frac{\sin C}{\sin B} = \frac{c}{b} = \frac{10}{9}; \text{ using cosine law } \cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{5}{16}$$

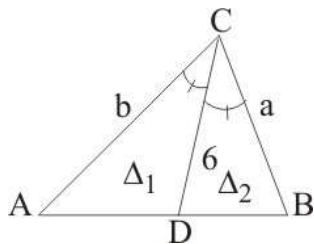
$$\therefore \frac{\tan C}{\sin B} = \frac{\sin C}{\sin B} = \frac{1}{\cos C} = \frac{10}{9} \cdot \frac{16}{5} = \frac{16}{5} = \frac{32}{9}$$

Q6. In a triangle ABC, CD is the bisector of the angle C. If $\cos \frac{C}{2}$ has the value $\frac{1}{3}$ and 1 (CD) = 6, then $\left(\frac{1}{a} + \frac{1}{b}\right)$ has the value equal to

- (A) $\frac{1}{9}$ (B) $\frac{1}{12}$ (C) $\frac{1}{6}$ (D) None

Ans. A

Sol.



$$\Delta = \Delta_1 + \Delta_2 = \frac{1}{2} ab \sin C = ab \sin \frac{C}{2} \cos \frac{C}{2}$$

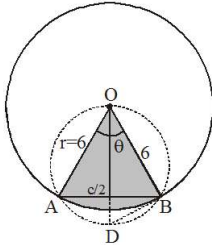
$$= \frac{1}{2} 6b \sin \frac{C}{2} + \frac{1}{2} 6a \sin \frac{C}{2} \Rightarrow \frac{C}{2} \sin \Rightarrow \frac{1}{a} + \frac{1}{b} = \frac{1}{9}$$

Q7. A sector OABO of central angle θ is constructed in a circle with centre O and of radius 6. The radius of the circle that is circumscribed about the triangle OAB, is

- (A) $6 \cos \frac{\theta}{2}$ (B) $6 \sec \frac{\theta}{2}$ (C) $3(\cos \frac{\theta}{2} + 2)$ (D) $3 \sec \frac{\theta}{2}$

Ans. D

Sol.



$$R = ; \Delta = \frac{1}{2} \cdot 6 \cdot 6 \sin \theta = 18 \sin \theta$$

$$a = b = 6$$

$$\sin \frac{\theta}{2} = \frac{c}{12} \Rightarrow c = 12 \frac{\theta}{2} \sin$$

$$R = \frac{36 \cdot 12 \sin(\theta/2)}{4 \cdot 18 \cdot \sin \theta} = 3 \sec \frac{\theta}{2}$$

$$\text{Alternatively : } OD^2 = 6^2 + x^2$$

$$\frac{x}{6} = \tan \frac{\theta}{2}$$

$$OD^2 = 6^2 + 6^2 \tan^2 = 6^2 \sec^2 \frac{\theta}{2} \Rightarrow OD = 6 \sec \frac{\theta}{2}$$

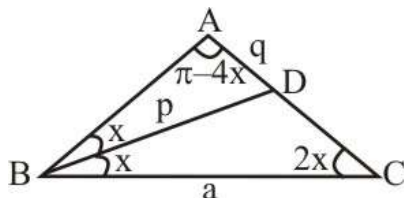
$$4r^2 = 36 \sec^2 \frac{\theta}{2} \Rightarrow r = 3 \frac{\theta}{2} \sec$$

Q8. Let triangle ABC be an isosceles triangle with $AB = AC$. Suppose that the angle bisector of its angle B meets the side AC at a point D and that $BC = BD + AD$. Measure of the angle A in degrees, is

- (A) 80 (B) 100 (C) 110 (D) 130

Ans. B

Sol.



$$BC = BD + AD$$

$$a = p + q$$

$$\text{or } \frac{a}{p} = 1 + \frac{q}{p}$$

using sine law in $\triangle BDC$ and in $\triangle ABD$

$$\frac{\sin 3x}{\sin 2x} = 1 + \frac{\sin x}{\sin 4x} \Rightarrow \frac{\sin 3x \cdot \sin 4x - \sin x \cdot \sin 2x}{\sin 2x \cdot \sin 4x} = 1$$

$$2 \sin 3x \cdot \sin 4x - 2 \sin x \cdot \sin 2x = 2 \sin 2x \cdot \sin 4x$$

$$(\cos x - \cos 7x) - (\cos x - 3x) = \cos 2x - \cos 6x$$

$$\cos 3x - 7x = \cos 2x - \cos 6x$$

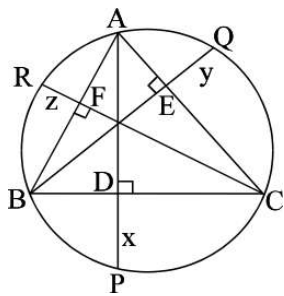
$$2 \sin 5x \cdot \sin 2x = 2 \sin 4x \cdot \sin 2x$$

$$\text{as } \sin 2x \neq 0$$

$$\text{hence } \sin 5x = \sin 4x$$

$$5x + 4x = 180^\circ \Rightarrow x = 20^\circ; \quad \text{hence } \angle A = 180^\circ - 80^\circ = 100^\circ$$

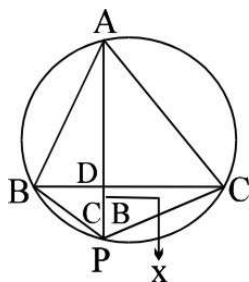
- Q9. As shown in the figure AD is the altitude on BC and AD produced meets the circumcircle of $\triangle ABC$ at P where $DP = x$. Similarly $EQ = y$ and $FR = z$. If a, b, c respectively denotes the sides BC, CA and AB then $\frac{a}{2x} + \frac{b}{2y} + \frac{c}{2z}$ has the value equal to



- (A) $\tan A + \tan B + \tan C$ (B) $\cot A + \cot B + \cot C$ (C) $\cos A + \cos B + \cos C$
 (D) $\operatorname{cosec} A + \operatorname{cosec} B + \operatorname{cosec} C$

Ans. A

Sol.



$$BD = x \tan C \text{ in } \triangle PDB$$

$$\text{and } DC = x \tan B \text{ for } \triangle PDC$$

$$\therefore BD + DC = a = x (\tan B + \tan C)$$

$$\frac{a}{x} = \tan B + \tan C$$

- Q10. In a $\triangle ABC$, the value of $\frac{a \cos A + b \cos B + c \cos C}{a + b + c}$ is equal to :

- (A) $\frac{r}{R}$ (B) $\frac{R}{2r}$ (C) $\frac{R}{r}$ (D) $\frac{2r}{R}$

Ans. A

Sol.
$$\text{LHS } \frac{R[\sin 2A + \sin 2B + \sin 2C]}{2R[\sin A + \sin B + \sin C]} = \frac{4\sin A \sin B \sin C}{2 \cdot 4 \cos \frac{B}{2} \cos \frac{C}{2}}$$

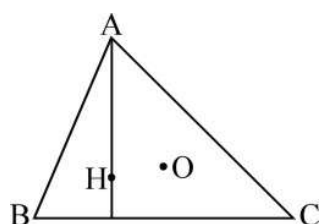
$$= \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \frac{r}{R}$$

Q11. ABC is an acute angled triangle with circumcentre 'O' orthocentre H. If $AO = AH$ then the measure of the angle A is

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Ans. C

Sol.



$$OA = HA$$

$R = 2R \cos A$ (distance of orthocentre from the vertex A is $2R \cos A$)

$$\Rightarrow \cos A = \frac{1}{2}$$

$$\Rightarrow A = \frac{\pi}{3}$$

Q12. In a ΔABC if $b + c = 3a$ then $\cot \frac{B}{2} \cdot \cot \frac{C}{2}$ has the value equal to :

- (A) 4 (B) 3 (C) 2 (D) 1

Ans. C

Sol.
$$\cot \frac{B}{2} \cdot \cot \frac{C}{2} = \frac{s(s-b)}{\Delta} \cdot \frac{s(s-c)}{\Delta} = \frac{s}{s-a} = \frac{2s}{2s-2a}$$

but given that $a + b + c = 4a \Rightarrow 2s = 4a$ Hence $\cot \frac{B}{2} \cdot \cot \frac{C}{2} = \frac{4a}{2a} = 2$

Q13. If A is the area and 2s the sum of the 3 sides of a triangle, then :

- (A) $A \leq \frac{s^2}{3\sqrt{3}}$ (B) $A = \frac{s^2}{2}$ (C) $A > \frac{s^2}{\sqrt{3}}$ (D) None

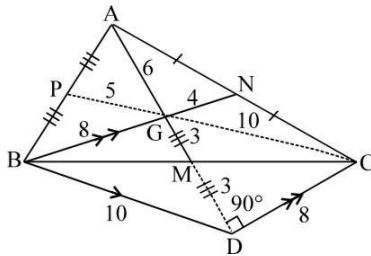
Ans. A

Sol. -

- Q14. The medians of a ΔABC are 9 cm, 12 cm and 15 cm respectively . Then the area of the triangle is
- (A) 96 sq cm (B) 84 sq cm (C) 72 sq cm (D) 60 sq cm

Ans. C

Sol.



Produce the median AM to D such that $GM = MD$. Join D to B and C. Now GBDC is parallelogram.

Note that the sides of the ΔGDC are 6, 8, 10 $\Rightarrow \angle GDC = 90^\circ$

$$\text{Area of } \Delta ADC = \frac{12 \cdot 8}{2} = 48$$

$$\text{Area of } \Delta MDC = \frac{3 \cdot 8}{2} = 12$$

- Q15. If l, m, n are the perpendiculars from the angular points of a ΔABC upon the opposite sides a, b, c respectively then, $\frac{bl}{c} + \frac{cm}{a} + \frac{an}{b}$ is equal to :

(A) $\frac{a^2 + b^2 + c^2}{2R}$ (B) $\frac{ab + bc + ca}{R}$ (C) $\frac{(a + b + c)^2}{4R}$ (D) $4R (1 + \cos A \cos B \cos C)$

Ans. D, A

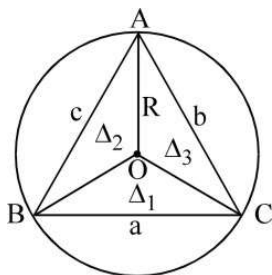
Sol. -

- Q16. If 'O' is the circumcentre of the ΔABC and R_1, R_2 and R_3 are the radii of the circumcircles of triangles OBC, OCA & OAB respectively then $\frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_3}$ has the value equal to

(A) $\frac{abc}{2R^3}$ (B) $\frac{R^3}{abc}$ (C) $\frac{4\Delta}{R^2}$ (D) $\frac{abc}{R^3}$

Ans. D, C

Sol.



$$\text{Using } R = \frac{abc}{4\Delta} \Rightarrow \frac{a}{R} = \frac{4\Delta}{bc}$$

$$\frac{a}{R_1} + \frac{b}{R_2} + \frac{c}{R_2} = \frac{4}{R^2}(\Delta_1 + \Delta_2 + \Delta_3) = \frac{4\Delta}{R^2}$$

Q17. In a ΔABC , following relations hold good . In which case(s) the triangle is a right angled triangle? (Assume all symbols have their usual meaning)

(A) $r_2 + r_3 = r_1 - r$ (B) $a^2 + b^2 + c^2 = 8R^2$

(C) If the diameter of an excircle be equal to the perimeter of the triangle.

(D) $2R = r_1 - r$

Ans. D, C, B, A

Sol. $\frac{\Delta}{s-b} + \frac{\Delta}{s-c} = \frac{\Delta}{s} \Rightarrow \frac{a}{(s-b)(s-c)} = \frac{a}{s(-a)}$ or

$$\frac{s(s-a)}{(s-b)(s-c)} = 1 \Rightarrow \cot \frac{A}{2} = 1 \Rightarrow A = 90^\circ$$

again $4R^2 (\sum \sin^2 A) = 8R^2 \Rightarrow \sin^2 A + \sin^2 B + \sin^2 C = 2$

$$\Rightarrow \cos A \cos B \cos C = 0 \Rightarrow A \text{ or } B \text{ or } C = 90^\circ$$

again $2r_1 = 2s \Rightarrow s \tan \frac{A}{2} = s \Rightarrow \tan \frac{A}{2} = 1 \Rightarrow A = 90^\circ$

again $2R = 4R [(\sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}) \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}]$

$$1 = 2 \sin \frac{A}{2} \cos \left(\frac{B+C}{2} \right) \text{ or } 1 = 2 \sin^2 A \Rightarrow \cos A = 0 \Rightarrow A = 90^\circ$$

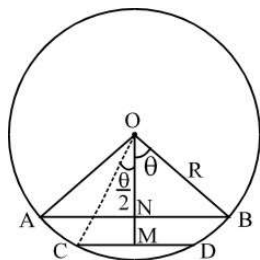
Q18. Two parallel chords are drawn on the same side of the centre of a circle of radius R . It is found that they subtend an angle of θ and 2θ at the centre of the circle . The perpendicular distance between the chords is

(A) $2R \sin \frac{3\theta}{2} \sin \frac{\theta}{2}$ (B) $\left(1 - \cos \frac{\theta}{2}\right) \left(1 + 2 \cos \frac{\theta}{2}\right) R$

(C) $\left(1 + \cos \frac{\theta}{2}\right) \left(1 - 2 \cos \frac{\theta}{2}\right) R$ (D) $2R \sin \frac{3\theta}{4} \sin \frac{\theta}{4}$

Ans. D, B

Sol.



$$OM = p_1 = R \cos \frac{\theta}{2} \quad ON = p_2 = R \cos \theta$$

$$MN = p_1 - p_2 = R \left(\frac{\cos \frac{\theta}{2} - \cos \theta}{2} \right)$$

$$= R 2 \sin \frac{3\theta}{4} \sin \frac{\theta}{4} \Rightarrow D$$

Again convert $\cos = 2 \cos^2 \frac{\theta}{2} - 1$ and factorise

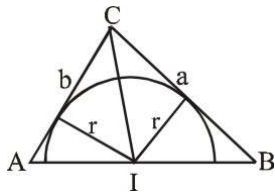
- Q19. In a $\triangle ABC$, a semicircle is inscribed, whose diameter lies on the side c . If x is the length of the angle bisector through angle C then the radius of the semicircle is

Where Δ is the area of the triangle ABC and 's' is semiperimeter.

(A) $\frac{abc}{4R^2(\sin A + \sin B)}$ (B) $\frac{\Delta}{x}$ (C) $x \sin \frac{C}{2}$ (D) $\frac{2\sqrt{s(s-a)(s-b)(s-c)}}{s}$

Ans. A, C

Sol.



$$\frac{1}{2}r a + \frac{1}{2}rb = \frac{1}{2}ab \sin C$$

$$r(a+b) = 2\Delta$$

$$r = \frac{\Delta}{a+b} \dots\dots\dots(1)$$

$$\therefore r = \frac{2abc}{4R(2R\sin A + 2R\sin B)} = \frac{abc}{4R^2(\sin A + \sin B)}$$

- Q20. In triangle ABC , let $a = 5$, $b = 4$ and $\cos(A - B) = \frac{31}{32}$, then which of the following statement(s) is (are) correct?

[Note : All symbols used have usual meaning in a triangle.]

- (A) The perimeter of triangle ABC equals $\frac{15}{2}$.
 (B) The radius of circle inscribed in triangle ABC equals $\frac{\sqrt{7}}{2}$.
 (C) The measure of angle C equals $\cos^{-1} \frac{1}{8}$.
 (D) The value of $R(b^2 \sin 2C + c^2 \sin 2B)$ equals 120.

Ans. C, B

Sol. -

- Q21. In which of the following situations, it is possible to have a triangle ABC ?
 (All symbols used have usual meaning in a triangle.)

(A) $(a + c - b)(a - c + b) = 4bc$ (B) $b^2 \sin 2C + c^2 \sin 2B = ab$

(C) $a = 3, b = 5, c = 7$ and $C = \frac{2\pi}{3}$ (D) $\cos\left(\frac{A-C}{2}\right) = \cos\left(\frac{A+C}{2}\right)$

Ans. C, B

Sol. -

Q22. In a triangle ABC, which of the following quantities denote the area of the triangle?

(A) $\frac{a^2 - b^2}{2} \left(\frac{\sin A \sin B}{\sin(A-B)} \right)$ (B) $\frac{r_1 r_2 r_3}{\sqrt{\sum r_1 r_2}}$ (C) $\frac{a^2 + b^2 + c^2}{\cot A + \cot B + \cot C}$
 (D) $r^2 \cot \frac{A}{2} \cdot \cot \frac{B}{2} \cdot \cot \frac{C}{2}$

Ans. D, B, A

Sol. -